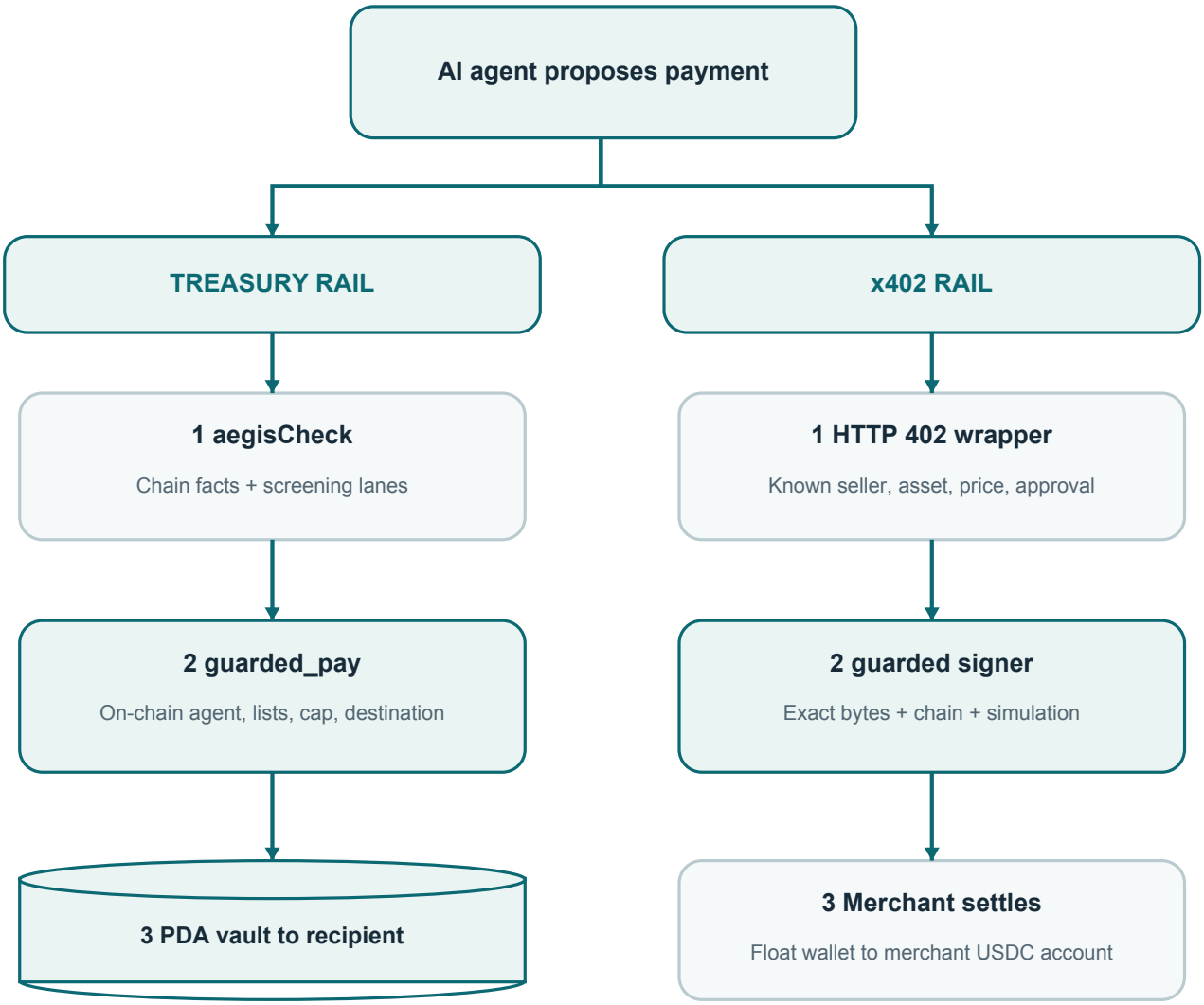
System overview

Two payment routes. Two hard gates. Same agent, same USDC, different enforcement points.



Treasury gate

Skipping the TypeScript check does not skip the Anchor guard. The vault cannot pay without guarded_pay.

x402 gate

The signer is binding only while the agent cannot read the float wallet's raw key.

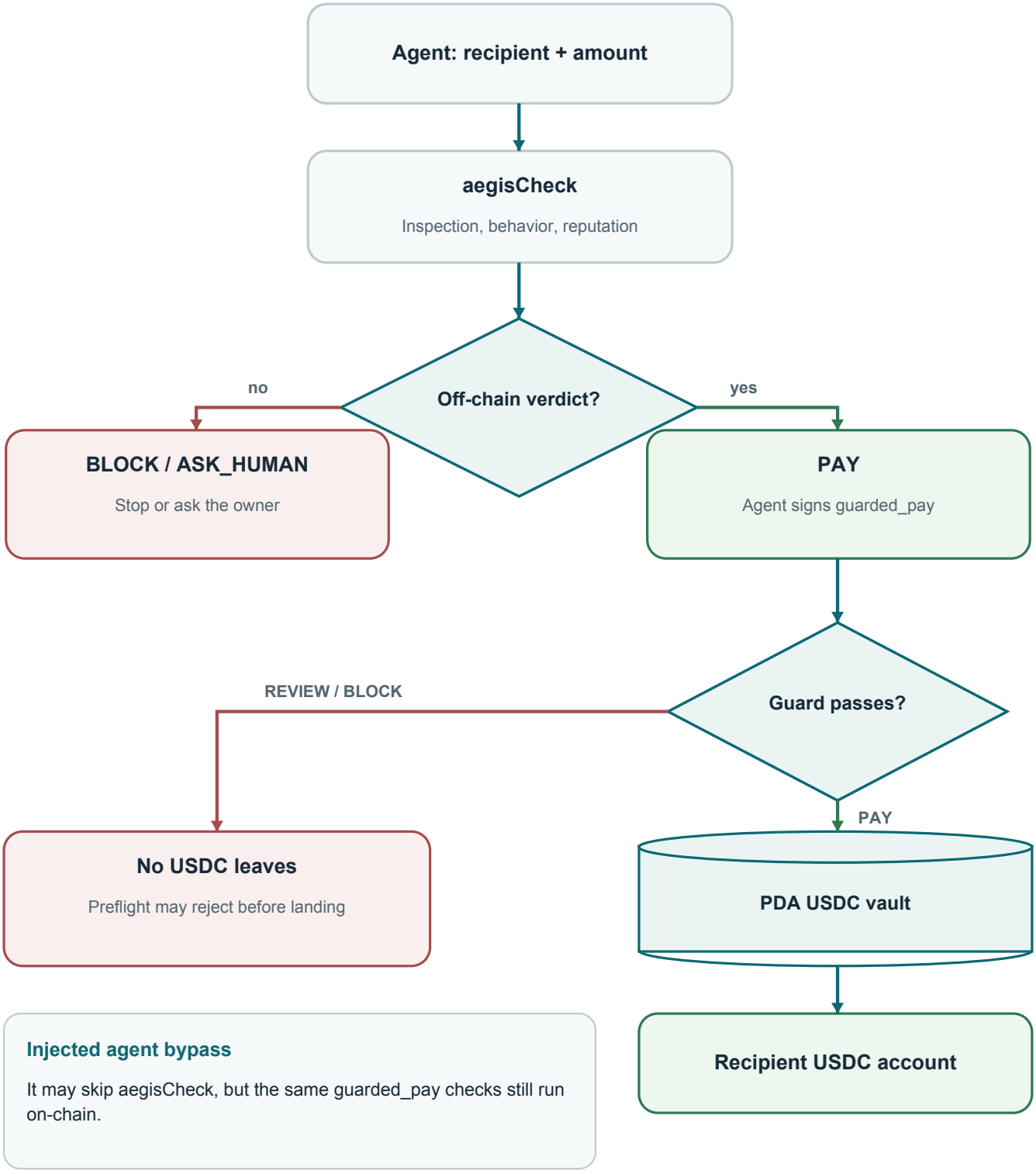
Shared link: a guarded treasury transfer refills the x402 float. The later merchant payment does not pass through the treasury guard.

KEY POINT

USDC moves only after the rail's hard gate passes. The two gates do different jobs.

Treasury rail

Normal payment flow: advice first, on-chain guard at transfer time.

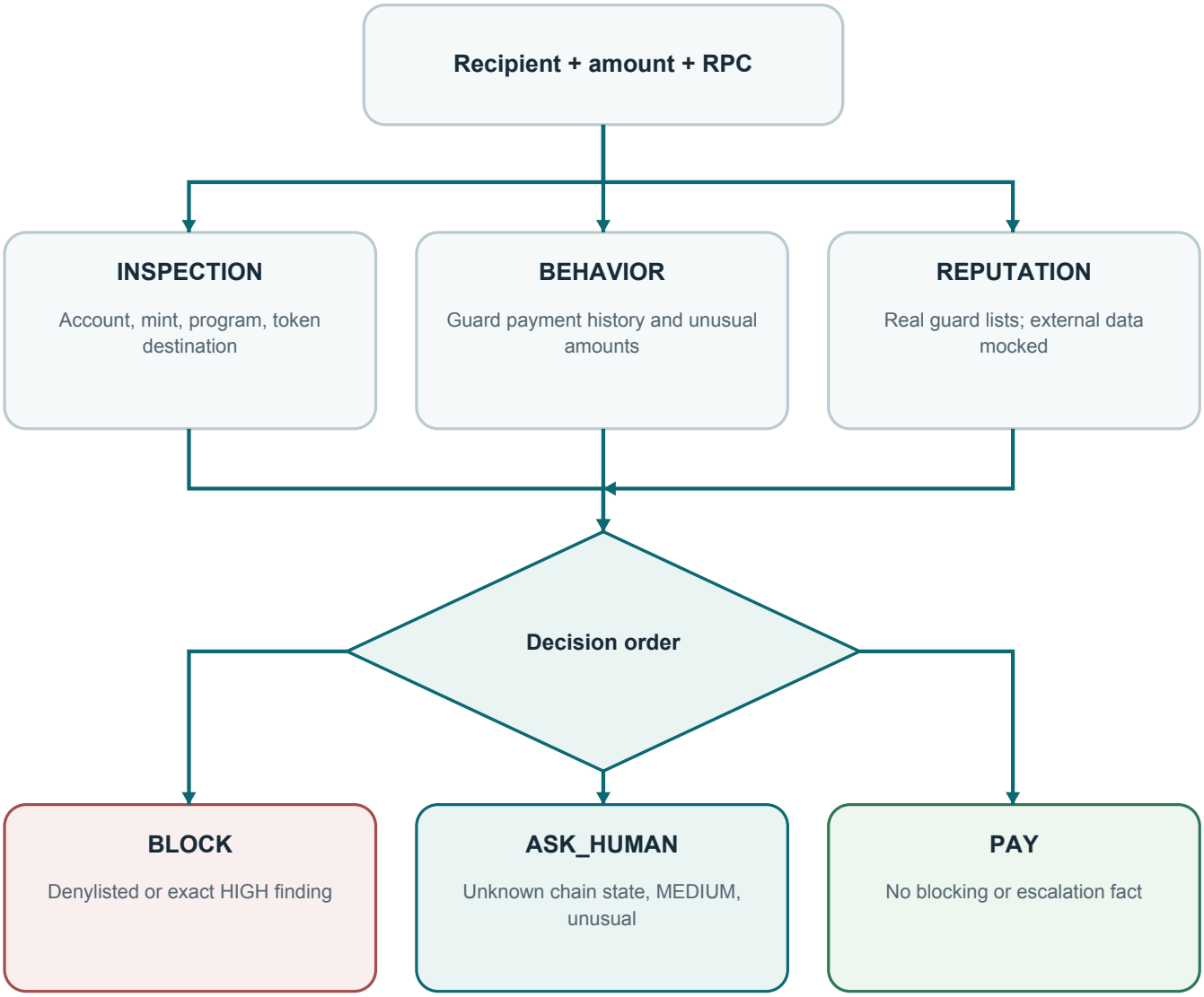


KEY POINT

The guard rechecks the agent, lists, daily cap, and destination inside the transfer.

Screening lanes

Three lanes read the facts. Deterministic order decides the answer.



Optional model

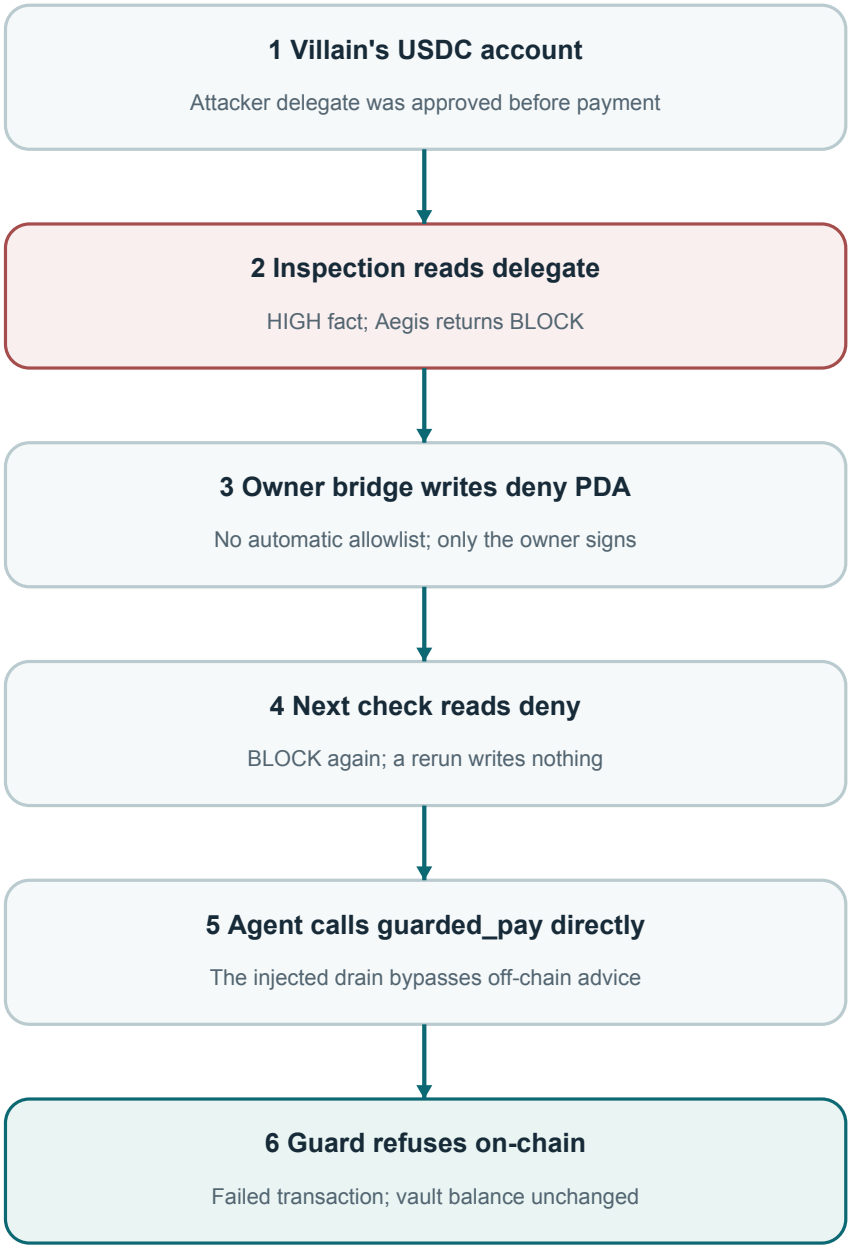
The optional model explains an escalation. It does not downgrade a HIGH fact into PAY. The demo skips the model.

KEY POINT

Real chain reads drive inspection and behavior; reputation's external source is mocked.

Bad recipient becomes a rule

A delegated token account is spotted, then the owner makes the block durable.



Important limit

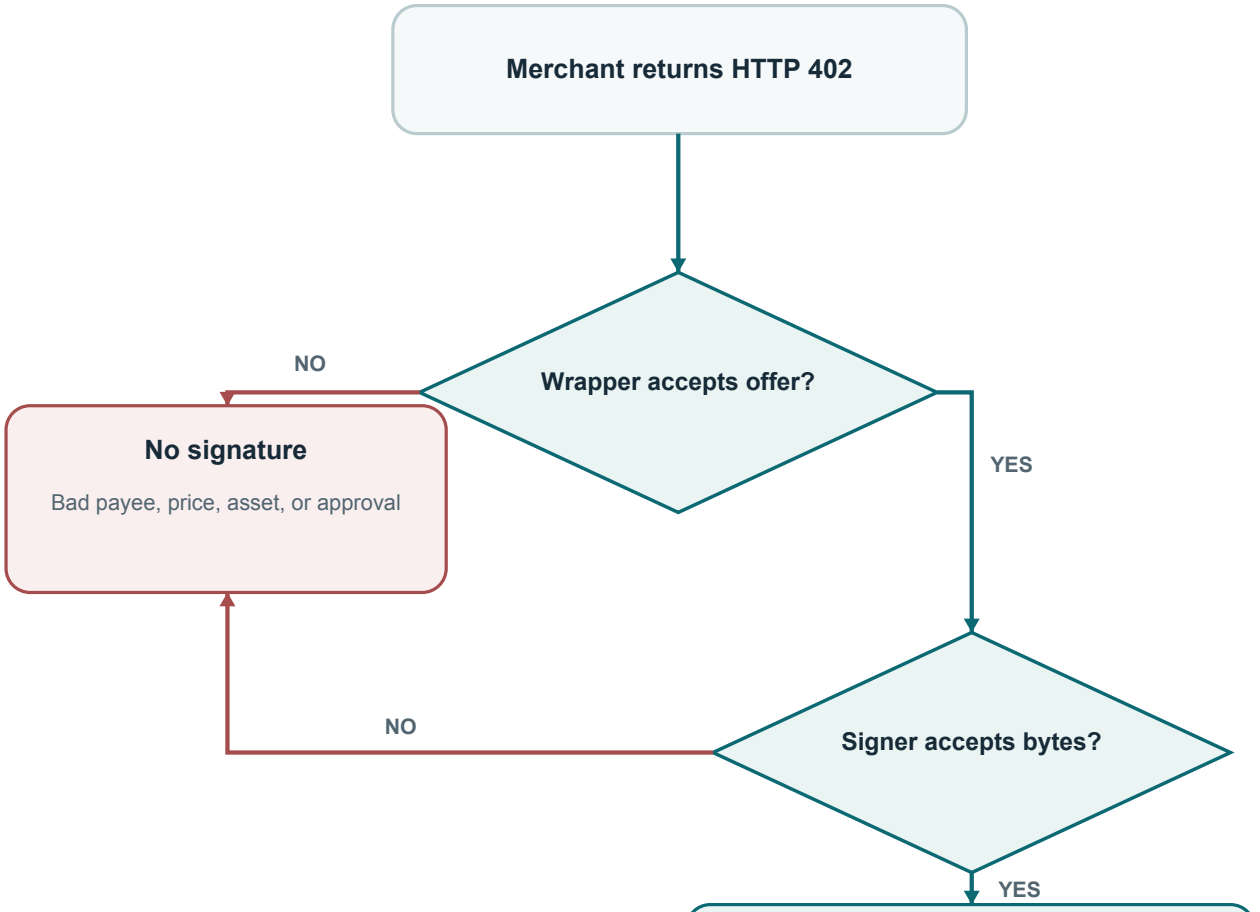
The check sees a delegate already present before payment. A delegate added after USDC arrives can still sweep those received tokens.

KEY POINT

The bridge records a deny rule; the guard enforces it even when the agent skips Aegis.

x402 merchant payment

The payment is stopped at the signer, before the float wallet gives a signature.



Product preparation

The merchant prepares response data before settlement, but its product callback must not email, trigger a webhook, or consume inventory then.

Bad-mint demo

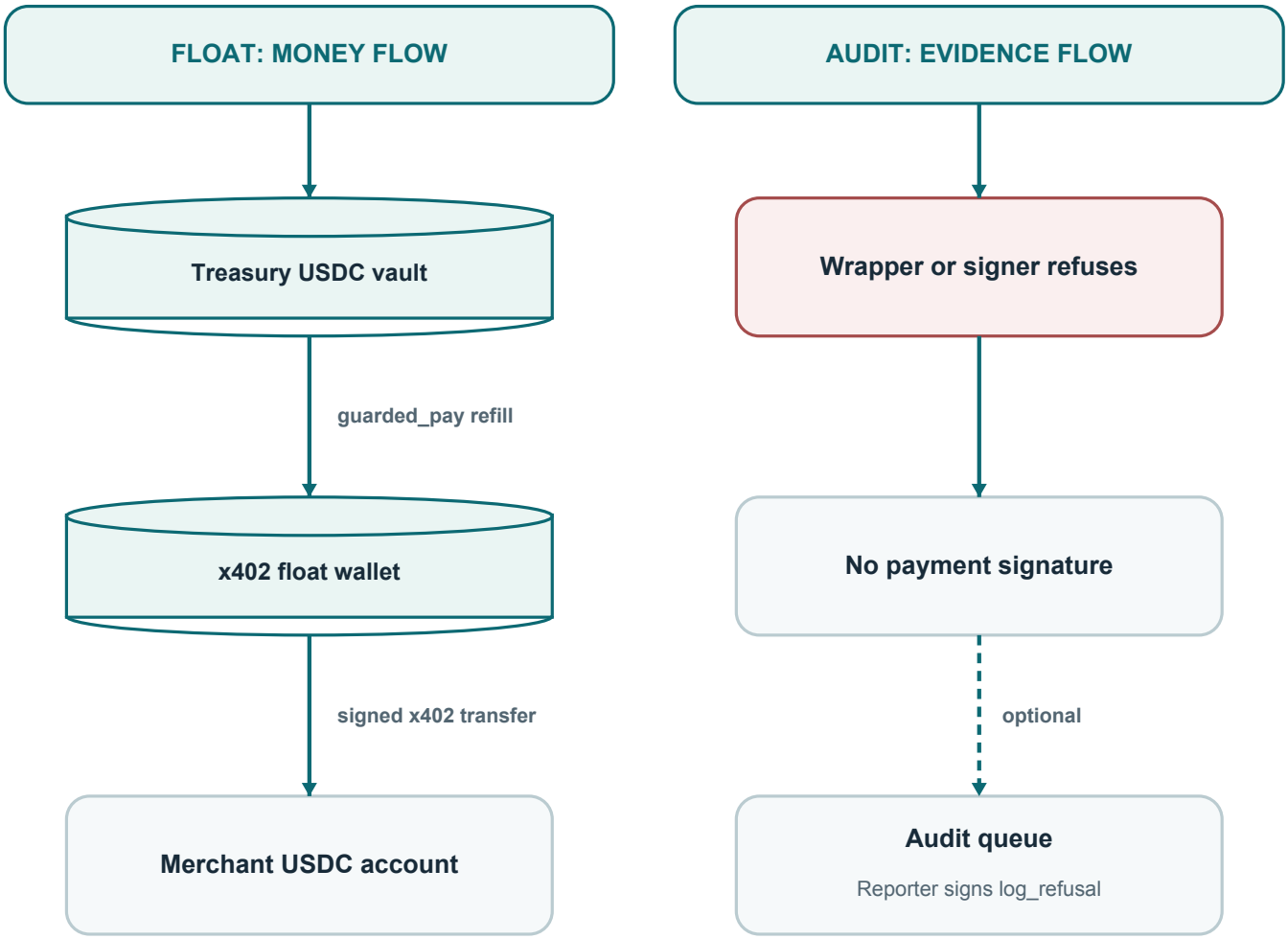
Token-2022 look-alike USDC is NO_PAYABLE_OPTION before signing.

KEY POINT

No valid signature means no x402 payment. The demo merchant is its own facilitator.

Float and refusal receipts

One flow moves USDC. The other records a refusal. These are separate flows.



Float limit

The wallet holds zero SOL. Refills use the guard's daily cap. Existing float is still spendable until its key is rotated or drained safely.

Receipt limit

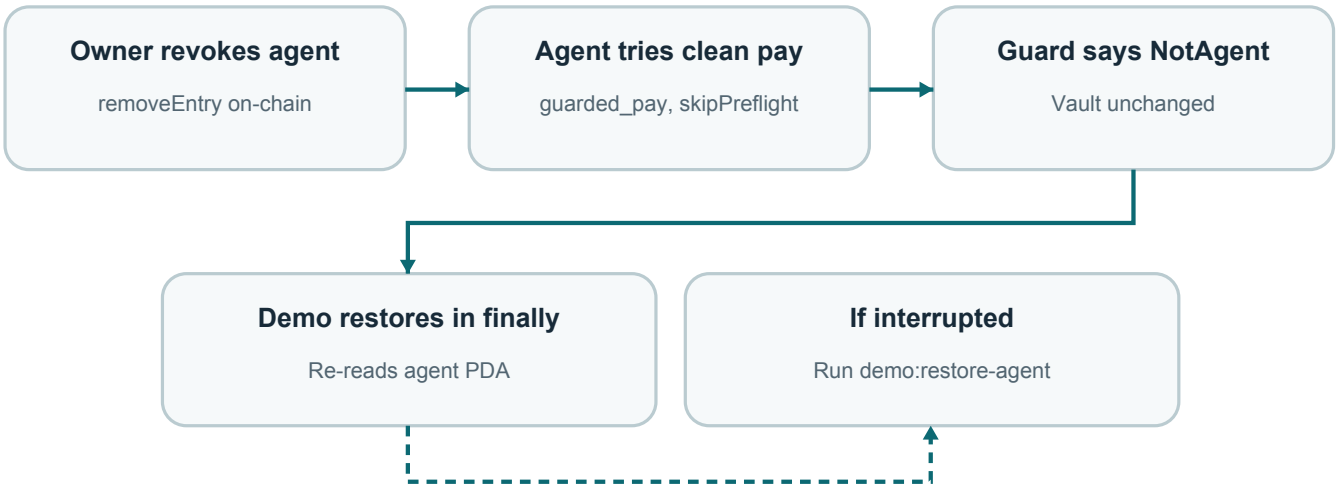
A receipt says a reporter logged a refusal. It is not the gate, and an empty log proves nothing. Writes are queued and capped per run.

KEY POINT

Guarded refill is the only path into the x402 float; the audit log is evidence, not enforcement.

Kill switch and reality check

Who can stop what, and which parts are real right now.



Trust boundary: owner and upgrade keys remain roots of trust. Revoking the treasury agent does not claw back the x402 float; stop refills and rotate that wallet key separately.

REAL VERSUS MOCKED

LOCALNET	Real Anchor transfers, failed transactions, HTTP 402, and receipts. USDC is a test mint.
DEVNET	Targeted by code; deployment, faucet funding, and demo run are owner steps.
FACILITATOR	Demo merchant settles itself. Independent facilitator and x402.org interop untested.
AI	Reputation external data mocked; LLM optional and skipped in the demo.

Known limits

x402.org and independent facilitator interop are untested. Merchant replay memory is in-process. A delegate added after receipt can still sweep USDC. Keep the raw x402 key away from the agent.

KEY POINT

Treasury kill switch is real. x402 float needs its own stop-refilling and key-rotation response.